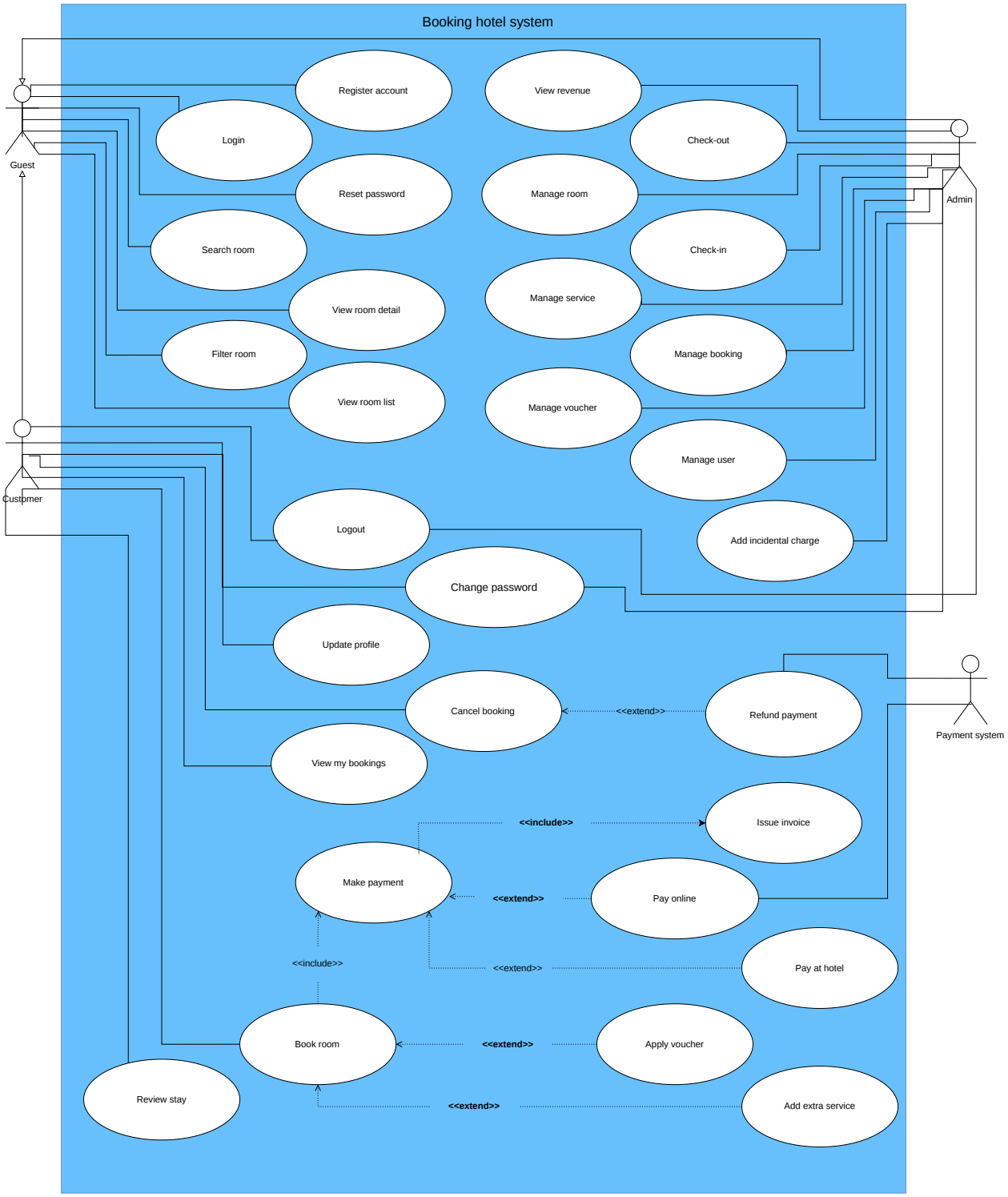
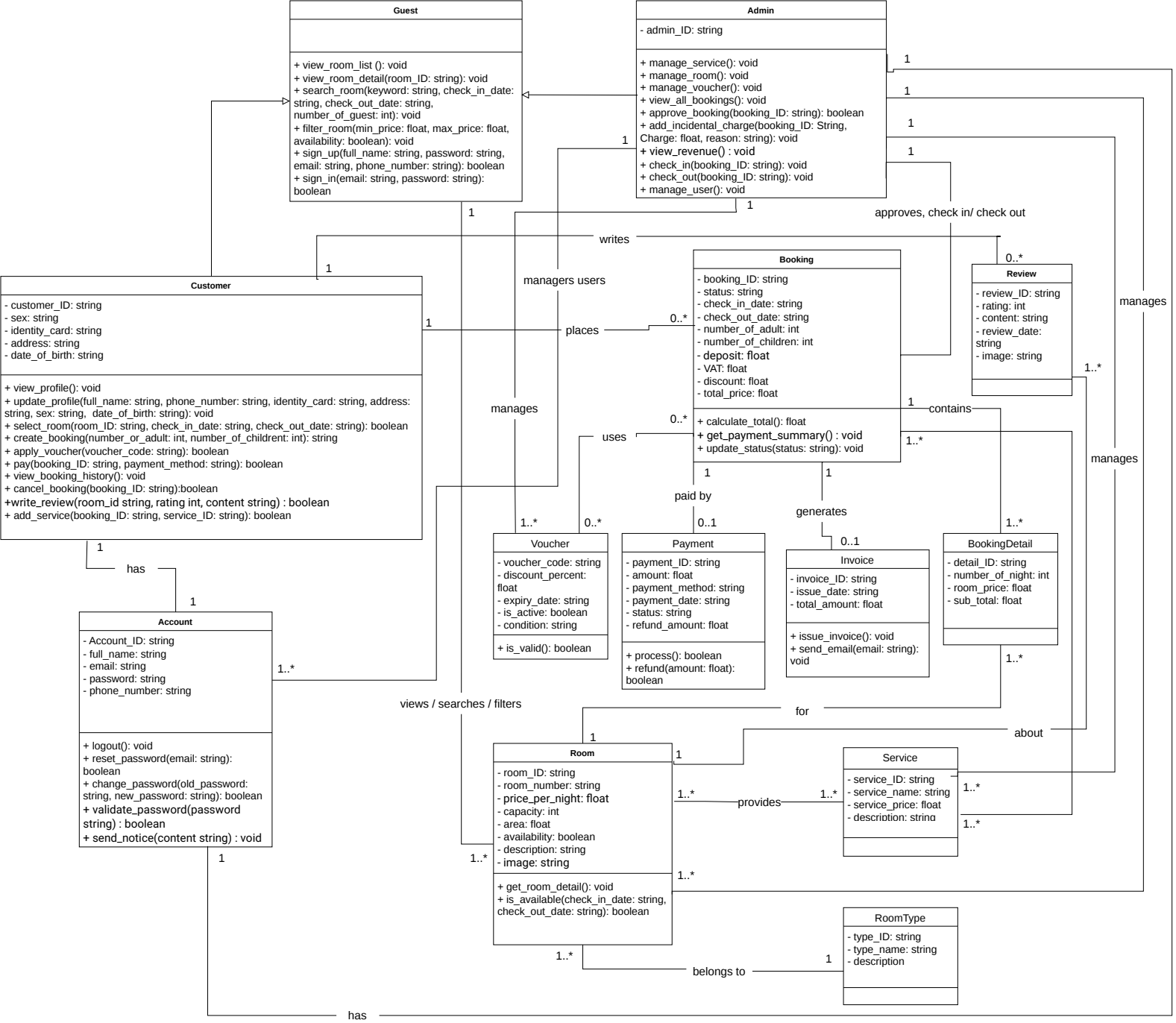
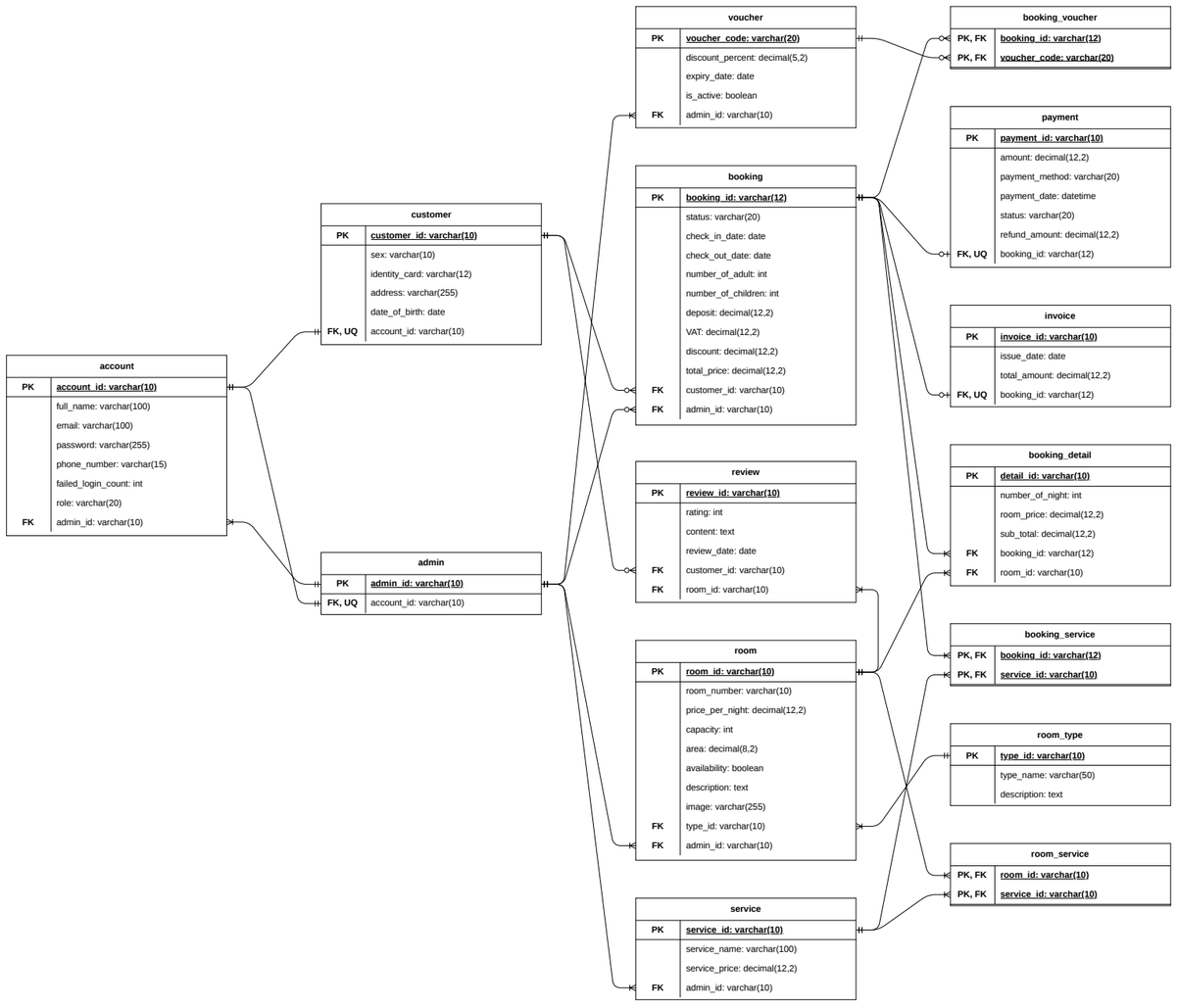

Booking hotel system
Guest
Customer
Admin
Payment system
Register account
Login
Reset password
Search room
View room detail
Filter room
View room list
View revenue
Check-out
Manage room
Check-in
Manage service
Manage booking
Manage voucher
Manage user
Add incidental charge
Logout
Change password
Update profile
Cancel booking
View my bookings
<<extend>>
Refund payment
<<include>>
Issue invoice
Make payment
<<extend>>
Pay online
<<include>>
<<extend>>
Pay at hotel
Book room
<<extend>>
Apply voucher
Review stay
<<extend>>
Add extra service


Guest
+ view_room_list (): void
+ view_room_detail(room_ID: string): void
+ search_room(keyword: string, check_in_date: string, check_out_date: string, number_of_guest: int): void
+ filter_room(min_price: float, max_price: float, availability: boolean): void
+ sign_up(full_name: string, password: string, email: string, phone_number: string): boolean
+ sign_in(email: string, password: string): boolean
Admin
- admin_ID: string
+ manage_service(): void
+ manage_room(): void
+ manage_voucher(): void
+ view_all_bookings(): void
+ approve_booking(booking_ID: string): boolean
+ add_incidental_charge(booking_ID: String, Charge: float, reason: string): void
+ view_revenue() : void
+ check_in(booking_ID: string): void
+ check_out(booking_ID: string): void
+ manage_user(): void
approves, check in/ check out
writes
Customer
- customer_ID: string
- sex: string
- identity_card: string
- address: string
- date_of_birth: string
+ view_profile(): void
+ update_profile(full_name: string, phone_number: string, identity_card: string, address: string, sex: string, date_of_birth: string): void
+ select_room(room_ID: string, check_in_date: string, check_out_date: string): boolean
+ create_booking(number_or_adult: int, number_of_children: int): string
+ apply_voucher(voucher_code: string): boolean
+ pay(booking_ID: string, payment_method: string): boolean
+ view_booking_history(): void
+ cancel_booking(booking_ID: string):boolean
+write_review(room_id string, rating int, content string) : boolean
+ add_service(booking_ID: string, service_ID: string): boolean
managers users
places
Booking
- booking_ID: string
- status: string
- check_in_date: string
- check_out_date: string
- number_of_adult: int
- number_of_children: int
- deposit: float
- VAT: float
- discount: float
- total_price: float
+ calculate_total(): float
+ get_payment_summary() : void
+ update_status(status: string): void
Review
- review_ID: string
- rating: int
- content: string
- review_date: string
- image: string
manages
contains
uses
paid by
generates
has
Account
- Account_ID: string
- full_name: string
- email: string
- password: string
- phone_number: string
+ logout(): void
+ reset_password(email: string): boolean
+ change_password(old_password: string, new_password: string): boolean
+ validate_password(password string) : boolean
+ send_notice(content string) : void
Voucher
- voucher_code: string
- discount_percent: float
- expiry_date: string
- is_active: boolean
- condition: string
+ is_valid(): boolean
Payment
- payment_ID: string
- amount: float
- payment_method: string
- payment_date: string
- status: string
- refund_amount: float
+ process(): boolean
+ refund(amount: float): boolean
Invoice
- invoice_ID: string
- issue_date: string
- total_amount: float
+ issue_invoice(): void
+ send_email(email: string): void
BookingDetail
- detail_ID: string
- number_of_night: int
- room_price: float
- sub_total: float
views / searches / filters
for
about
Room
- room_ID: string
- room_number: string
- price_per_night: float
- capacity: int
- area: float
- availability: boolean
- description: string
- image: string
+ get_room_detail(): void
+ is_available(check_in_date: string, check_out_date: string): boolean
provides
Service
- service_ID: string
- service_name: string
- service_price: float
- description: string
belongs to
RoomType
- type_ID: string
- type_name: string
- description
has


account
PK account_id: varchar(10)
full_name: varchar(100)
email: varchar(100)
password: varchar(255)
phone_number: varchar(15)
failed_login_count: int
role: varchar(20)
FK admin_id: varchar(10)
customer
PK customer_id: varchar(10)
sex: varchar(10)
identity_card: varchar(12)
address: varchar(255)
date_of_birth: date
FK, UQ account_id: varchar(10)
admin
PK admin_id: varchar(10)
FK, UQ account_id: varchar(10)
voucher
PK voucher_code: varchar(20)
discount_percent: decimal(5,2)
expiry_date: date
is_active: boolean
FK admin_id: varchar(10)
booking
PK booking_id: varchar(12)
status: varchar(20)
check_in_date: date
check_out_date: date
number_of_adult: int
number_of_children: int
deposit: decimal(12,2)
VAT: decimal(12,2)
discount: decimal(12,2)
total_price: decimal(12,2)
FK customer_id: varchar(10)
FK admin_id: varchar(10)
review
PK review_id: varchar(10)
rating: int
content: text
review_date: date
FK customer_id: varchar(10)
FK room_id: varchar(10)
room
PK room_id: varchar(10)
room_number: varchar(10)
price_per_night: decimal(12,2)
capacity: int
area: decimal(8,2)
availability: boolean
description: text
image: varchar(255)
FK type_id: varchar(10)
FK admin_id: varchar(10)
service
PK service_id: varchar(10)
service_name: varchar(100)
service_price: decimal(12,2)
FK admin_id: varchar(10)
booking_voucher
PK, FK booking_id: varchar(12)
PK, FK voucher_code: varchar(20)
payment
PK payment_id: varchar(10)
amount: decimal(12,2)
payment_method: varchar(20)
payment_date: datetime
status: varchar(20)
refund_amount: decimal(12,2)
FK, UQ booking_id: varchar(12)
invoice
PK invoice_id: varchar(10)
issue_date: date
total_amount: decimal(12,2)
FK, UQ booking_id: varchar(12)
booking_detail
PK detail_id: varchar(10)
number_of_night: int
room_price: decimal(12,2)
sub_total: decimal(12,2)
FK booking_id: varchar(12)
FK room_id: varchar(10)
booking_service
PK, FK booking_id: varchar(12)
PK, FK service_id: varchar(10)
room_type
PK type_id: varchar(10)
type_name: varchar(50)
description: text
room_service
PK, FK room_id: varchar(10)
PK, FK service_id: varchar(10)